PDFM algorithm

1 Initialization — Once

Train regular FM

$\theta \leftarrow \theta - \eta^{(\theta)} \nabla_\theta \mathcal{L}_{\mathrm{FM}}(\theta)$

until convergence.

Initialize classifier f_ϕ

Label rollouts from the FM model.

Train f_ϕ to convergence.

First dual update

$\lambda'(1) = [\eta_0 (b - \widehat{P}_{u_\theta}(\mathcal{C}))]_+.$

2 Primal updates — Fix λ'

Generate trajectories and query membership

$x_0 \sim P_0, \quad x_{t+\Delta t} = x_t + u_\theta(x_t, t)\Delta t \quad \rightarrow \quad x_{\Delta t}, \ldots, x_{N\Delta t}, \qquad y = \mathbf{1}_{\mathcal{C}}(x_{N\Delta t}).$

Pair each (state, time) $(x_{i\Delta t}, i\Delta t)$ with the same terminal label y.

Update generator u_θ

$\theta \leftarrow \theta - \eta^{(\theta)} \nabla_\theta \left[\mathcal{L}_{\mathrm{FM}}(\theta) + \lambda' \big(b - P_{u_\theta}(\mathcal{C}) \big) \right],$

with P_{u_θ}'s gradients through f_ϕ.

Update classifier f_ϕ

Fit $f_\phi(x_{i\Delta t}, i\Delta t)$ to y.

Refresh trajectories and alternate updates until primal convergence.

Repeat until Convergence

3 Dual updates — Target b

Estimate satisfaction rate

$\widehat{P}_{u_\theta}(\mathcal{C}) = \frac{1}{M} \sum_{j=1}^{M} y_j$

Fresh terminal labels: $y_j = \mathbf{1}_{\mathcal{C}}(x^{(j)}_{N\Delta t})$.

Adjust dual variable λ'

$\lambda' \leftarrow [\lambda' + \eta_i (b - \widehat{P}_{u_\theta}(\mathcal{C}))]_+$

Trained u_θ

Output — After training

Regular FM Euler sampling

$x_0 \sim P_0, \quad x_{t+\Delta t} = x_t + u_\theta(x_t, t)\Delta t$

until $t = 1$.

Generated sample

$x_1 \sim P_{u_\theta}$